

AURA: Unsupervised Autoencoder-Based Detection of Bank Account Opening Fraud

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Abstract—Financial institutions lose hundreds of billions of dollars annually to fraud, and supervised detectors degrade against novel attack patterns and demand continuously relabelled data. We investigate a label-free alternative: an autoencoder trained only on legitimate bank-account-opening applications, using reconstruction error as an anomaly score. Using the one-million-record *Base* variant of the Bank Account Fraud (BAF) benchmark, in which frauds constitute just 1.10% of records, we build a leakage-free pipeline—splitting before fitting any transformation, correctly handling the dataset’s -1 missing-value sentinels and its true categorical cardinalities, and scaling with outlier clipping. The threshold is selected on a validation set and the model is evaluated on a held-out test set with imbalance-aware metrics. The autoencoder attains a test ROC-AUC of 0.592 and average precision (PR-AUC) of 0.016, exceeding an Isolation Forest baseline (0.553; 0.014) and the random-classifier prevalence floor (0.011). While the absolute separation is modest—as expected for a purely unsupervised model on this deliberately hard, biased benchmark—the results are honest and reproducible, and position the autoencoder as a label-free first-line filter that narrows the population passed to downstream supervised models or human analysts. We report accuracy only to demonstrate why it is misleading here: predicting “never fraud” already yields 98.9% accuracy.

Index Terms—anomaly detection, autoencoder, fraud detection, unsupervised learning, class imbalance, precision–recall

I. INTRODUCTION

According to Nasdaq’s Global Financial Crime Report [2], financial institutions lose on the order of hundreds of billions of dollars each year to fraud. Traditional rule-based and supervised detection systems struggle with three recurring problems: they respond poorly to zero-day attack patterns not present in historical labels, they generate excessive false positives, and they require constant manual relabelling as fraud tactics evolve. Modern threats—account-takeover using stolen credentials, synthetic-identity fraud, and layered money laundering—are specifically engineered to resemble normal behaviour, further eroding the value of signature-based rules.

This motivates *unsupervised* anomaly detection, which requires no fraud labels at training time. We study an autoencoder [3], [4]: a neural network trained to compress and reconstruct its input. Trained exclusively on legitimate transactions, it learns a compact representation of “normal” behaviour; inputs it cannot reconstruct well—quantified by the reconstruction error—are flagged as candidate anomalies.

Because fraud labels are rare, noisy, or unavailable in practice, a detector that needs none is attractive as a first line of defence.

We name the system AURA. Our contributions are: (i) a corrected, leakage-free pipeline for the BAF benchmark that fixes several data-handling errors common to naive implementations; (ii) an honest, imbalance-aware evaluation using precision–recall analysis rather than accuracy; and (iii) a controlled comparison against an Isolation Forest baseline and the theoretical random floor. We deliberately foreground the limitations of unsupervised detection on this dataset rather than overstating results.

II. DATASET

We use the *Base* variant of the Bank Account Fraud (BAF) dataset suite [1], published at NeurIPS 2022. It is a large, realistic *synthetic* tabular dataset of online bank-account-opening applications, released precisely to support reproducible fraud-detection research without exposing customer data. The *Base* table contains 1,000,000 records and 32 columns, including the binary target `fraud_bool`. Fraud is highly imbalanced: only 11,029 records (1.10%) are fraudulent.

The dataset has three properties that must be handled carefully and that naive pipelines routinely mishandle. First, several numeric fields encode *missing* values as -1 rather than NaN; for example 71.3% of `prev_address_months_count` and 25.4% of `bank_months_count` are -1 . Treating -1 as a genuine magnitude distorts scaling and distance-based anomaly scores. Second, two categorical fields have higher cardinality than is often assumed: `employment_status` and `housing_status` each contain *seven* categories (CA–CG and BA–BG), not five. Third, `device_fraud_count` is constant across all rows and carries no information.

III. METHODOLOGY

A. Preprocessing

All preprocessing is fit on the training partition only, and specifically on its *normal* subset, to prevent information from the validation and test sets (or from fraud examples) leaking into the learned representation. The steps are:

- **Drop constant features.** `device_fraud_count` is removed.

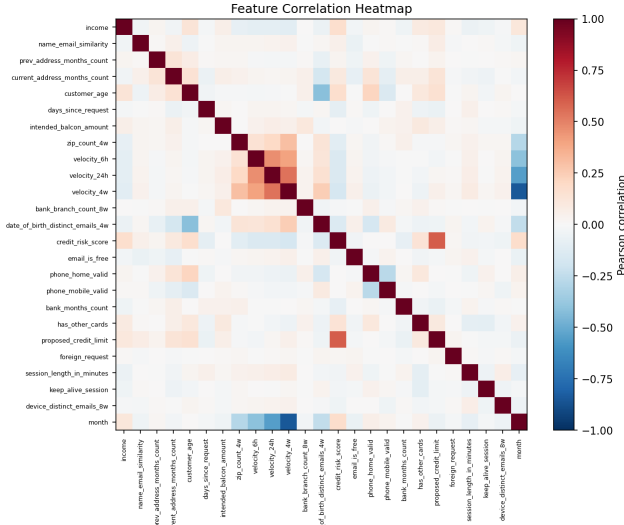


Fig. 1. Pearson correlation heatmap of the numeric features (with -1 sentinels treated as missing). The strongest correlation is `velocity_4w-month` (-0.85); no pair is redundant enough to drop.

- **Missing-value sentinels.** For the six columns using the -1 convention we add a binary *missing-indicator* feature and replace -1 with the median of the training-normal column (median imputation, fit on train-normal only). This preserves the signal that a value was missing—often itself predictive—without injecting a spurious magnitude.
- **Outlier clipping.** Continuous features are clipped to their 1st and 99th training-normal percentiles, limiting the influence of extreme values on a reconstruction objective.
- **Min-Max scaling.** Continuous features are scaled to $[0, 1]$. Under a mean-squared-error objective, unscaled features with large ranges (e.g. `velocity_6h`, which reaches $\sim 16,700$) would otherwise dominate the loss.
- **Categorical encoding.** The five nominal categoricals are one-hot encoded using category levels fixed from the training data, avoiding the ordinal-integer mapping used previously, which imposes a false ordering on unordered categories.

After encoding, the input dimensionality is 57. A correlation analysis (Fig. 1) shows the strongest pairwise Pearson correlation is between `velocity_4w` and `month` (-0.85), an artefact of the temporal structure; no continuous–continuous pair is redundant enough to warrant removal, so all features are retained.

B. Data Partitioning

The data are split 50/25/25 into training, validation, and test sets, stratified on `fraud_bool` so that the 1.10% prevalence is preserved in each partition. The autoencoder is trained only on the *normal* rows of the training partition (494,485 records). The validation set (with labels) is used solely to select the decision threshold; the test set is used once, for final reporting.

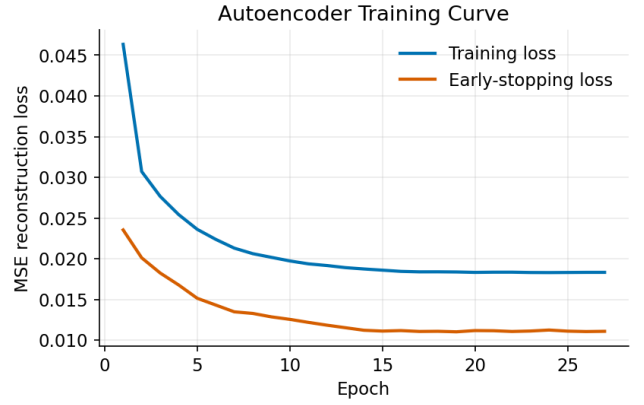


Fig. 2. Training and early-stopping reconstruction loss. Training halts at epoch 27 when the held-out normal loss stops improving.

C. Autoencoder Architecture

The model is a symmetric feed-forward autoencoder. The encoder maps the 57-dimensional input through hidden layers of 64 and 32 units to a 16-dimensional bottleneck; the decoder mirrors this back to 57 outputs. Each hidden layer uses batch normalisation, a ReLU activation, and dropout ($p = 0.1$); weights are initialised with Xavier uniform initialisation. The output layer uses a sigmoid activation, matching the $[0, 1]$ range of the scaled inputs. This configuration was selected by a validation-set sweep over bottleneck width $\{4, 8, 16, 24\}$, encoder depth, and dropout.

The training objective is the mean-squared reconstruction error

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2, \quad (1)$$

where $\mathbf{x}_i$ is an input and $\hat{\mathbf{x}}_i$ its reconstruction. At inference the per-sample reconstruction error serves as the anomaly score $s(\mathbf{x}) = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$, and a sample is flagged as fraudulent when $s(\mathbf{x}) > \theta$.

D. Training and Thresholding

The model is trained with the Adam optimiser [6] (learning rate 10^{-3} , weight decay 10^{-5}), batch size 256, for up to 60 epochs. Ten percent of the training-normal data is held out to monitor reconstruction loss, and training stops early when this loss fails to improve for eight epochs; in our run it halted at epoch 27 (Fig. 2). The decision threshold θ is chosen on the validation set as the value maximising the F_1 score of the fraud class—never on the test set.

E. Baseline

For context we compare against an Isolation Forest [5] (100 trees), a standard unsupervised anomaly detector, fit on the same training-normal data with its threshold selected on validation in the same way.

TABLE I
TEST-SET PERFORMANCE (250,000 RECORDS, 1.10% FRAUD).

Model	ROC-AUC	PR-AUC	Prec.	Recall	F_1
Autoencoder	0.592	0.016	0.022	0.153	0.038
Isolation Forest	0.553	0.014	0.016	0.135	0.029
Random baseline	0.500	0.011	0.011	—	—

IV. EXPERIMENTAL SETUP

The pipeline is implemented in PyTorch and scikit-learn. All randomness (NumPy, Python, PyTorch) is seeded (42) for reproducibility. The model is small ($\sim 13k$ parameters) and trains in under three minutes on CPU and in seconds on a commodity GPU; the accompanying notebook auto-detects and uses CUDA when available. We report ROC-AUC, average precision (PR-AUC), and precision/recall/ F_1 at the selected threshold. For a problem with 1.10% prevalence, average precision is the primary metric: the random-classifier baseline for PR-AUC equals the prevalence (0.011), whereas ROC-AUC’s random baseline is 0.5 and can appear deceptively healthy.

V. RESULTS

Table I summarises test-set performance. The autoencoder achieves a ROC-AUC of 0.592 and an average precision of 0.016, outperforming the Isolation Forest (0.553 and 0.014) and clearing the random floors on both metrics. At the validation-selected threshold ($\theta = 0.0248$) it recalls 15.3% of frauds at 2.2% precision.

The ROC and precision–recall curves (Fig. 3) show the autoencoder dominating both the baseline and the random reference across most operating points. The reconstruction-error distribution (Fig. 4, left) makes the difficulty explicit: normal and fraudulent errors overlap heavily, but fraud exhibits a consistently heavier right tail, which is the signal the anomaly score exploits. The confusion matrix (Fig. 4, right) shows the corresponding operating point.

We deliberately report accuracy last: at this threshold overall accuracy is 91.5%, and a trivial “never fraud” classifier scores 98.9%. Accuracy is therefore uninformative here and is reported only to caution against it.

VI. DISCUSSION

The autoencoder provides a real but modest improvement over random and over a standard baseline. This ceiling is consistent with the nature of the task: an unsupervised model never observes a fraud label, and the BAF benchmark was explicitly constructed to be biased, imbalanced, and difficult, with published *supervised* baselines reaching ROC-AUC around 0.85 [1]. The gap between 0.59 (unsupervised) and 0.85 (supervised) is the price of requiring no labels.

The practical value of AURA is therefore not as a standalone classifier but as a label-free *first-line filter*. Ranking transactions by reconstruction error lets an institution concentrate limited review capacity on the highest-risk tail, and the model requires no labelled fraud to deploy or to adapt as “normal”

behaviour drifts. Its main limitations are the heavy overlap between legitimate-but-unusual behaviour and genuine fraud, and the limited interpretability of a reconstruction score—a barrier in regulated settings where adverse decisions must be explained.

VII. CONCLUSION AND FUTURE WORK

We presented AURA, an unsupervised autoencoder for bank-account-opening fraud detection, evaluated honestly on the BAF benchmark with a leakage-free pipeline and imbalance-aware metrics. The model attains a test ROC-AUC of 0.592 and PR-AUC of 0.016, beating an Isolation Forest baseline while requiring no fraud labels. Future work includes coupling the reconstruction score with a lightweight supervised head (semi-supervised detection), sequence-aware architectures (LSTM or transformer autoencoders) to capture temporal patterns, and per-feature error attribution to improve explainability. Combining the autoencoder’s label-free adaptability with a supervised second stage is the most promising route to closing the gap with fully supervised detectors while retaining robustness to novel fraud.

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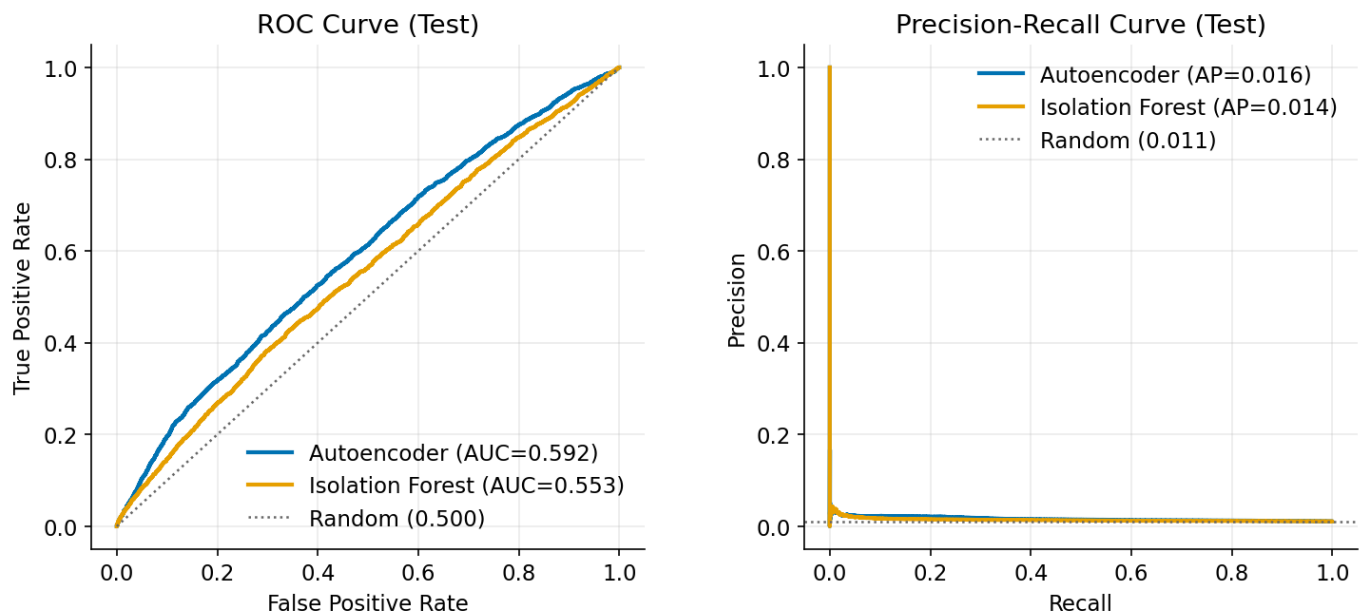


Fig. 3. Test-set ROC (left) and precision-recall (right) curves. The autoencoder (blue) exceeds the Isolation Forest (orange) and the random references (dotted). The precision-recall view is the honest one for 1.10% prevalence, where the random average-precision floor is only 0.011.

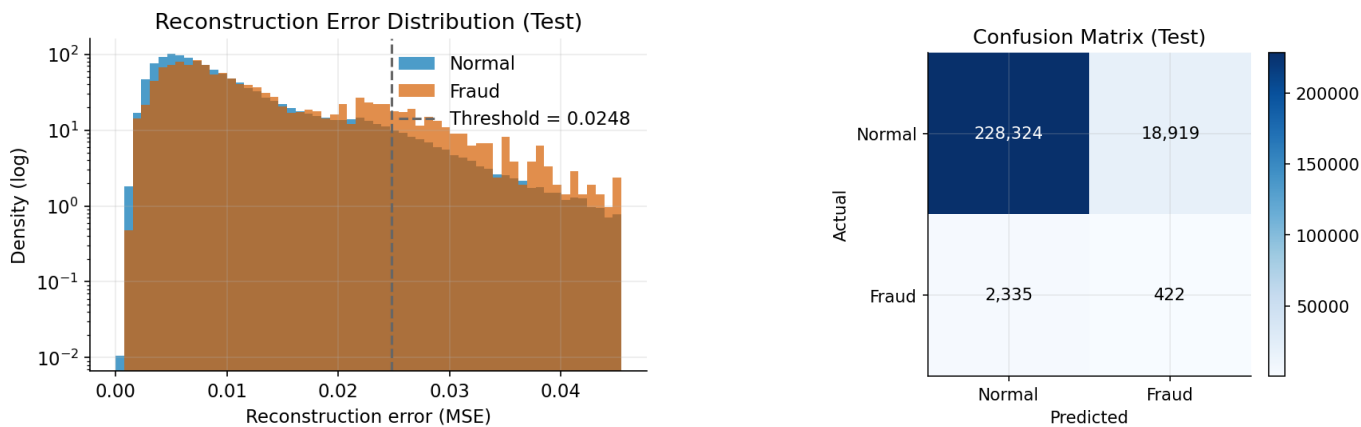


Fig. 4. Left: reconstruction-error distribution on the test set (log density); fraud has a heavier right tail. Right: confusion matrix at the validation-selected threshold.